

nology and the human body. Hypothetical technologies such as digital tattoo interfaces can combine body-modification aesthetics with interactivity and functionality.

By defense forces. Military departments have been researching utilisation of cyborg animals for tactical advantage purposes. Defense Advanced Research Projects Agency (DARPA), an agency of the US Department of Defense, is in the process of developing cyborg insects to transmit data from sensors implanted into an insect during the pupal stage. The insect's motion is controlled from a micro-electro-mechanical system (MEMS) and it can be used to survey an environment or detect explosives and gas.

In 2006, researchers at Cornell University, USA, invented a surgical procedure to implant artificial structures into insects during their metamorphic development. The first insect cyborgs, with integrated electronics in their thorax to be demonstrated, were moths. The initial success of the technique led to further research and the birth of a programme called Hybrid-Insect-MEMS (HI-MEMS).

The objective of HI-MEMS is to develop tightly-coupled machine-insect interfaces by placing micro-mechanical systems inside the insects during the early stages of metamorphosis. In time to come, researchers plan to develop HI-MEMS for dragonflies, bees, rats and pigeons.

For HI-MEMS cybernetic bug to be considered successful, it should be able to fly to a height of at least 100m (330-feet), have the ability to be guided by a computer into a controlled landing within 5m (16-feet) of a particular end point, and once it has landed, the bug should be able to be made stationary.

On similar lines, DARPA is developing a neural implant to remotely control the movement of sharks,



Fig. 5: Cyborg insects developed by DARPA (Image courtesy: omnibusintelligence.blogspot.in)



Fig. 6: In the future, perhaps we will be simply able to insert artificial eyes (Image courtesy: weheartit.com)

since their unique senses can be exploited to provide data feedback in relation to the location of enemy ships or underwater explosives.

In space exploration. Space exploration is risky for human beings and implementation of various cyborg technologies can prove to be very useful and help in risk mitigation. It was Stephen Hawking who said "Life on Earth is at the ever-increasing risk of being wiped out by a disaster such as sudden global warming or nuclear war. I think the human race has no future if it does not go into space."

The various challenges associated with space travel in terms of lack of oxygen, food, gravity, exercise and so on can imply we will take several more decades before we can start visiting and even living on various planets—perhaps planet Mars to start with.

While space travel and exploration has many taxing effects on the human body, the main is the

biological need for oxygen. If this aspect could be taken care of, then space exploration could be totally transformed.

Two eminent scientists, Manfred E. Clynes and Nathan S. Kline, have proposed a solution. They have put forward the application of an inverse fuel cell that has the capability to break down carbon dioxide into its components, eliminate carbon and re-circulate oxygen.

Another inhibiting factor in space travel is exposure to radiation. On an average, a human being is exposed to approximately 0.30 rem of radiation on Earth. On the other hand, an astronaut aboard the International Space Station for about a three-month time period is exposed to a radiation level as high as 9 rem. To address this problem, Clynes and Kline theorised a cyborg containing a sensor that would detect radiation levels and a rose-osmotic pump that would automatically inject protective medicines in the required quantity. Lab experiments of injecting these into monkeys as protective medicines have shown positive results in increasing radiation resistance.

The future

It is imagined that cyborg technology will form a part of the future human evolution. Today we already have MindMesh, an EEG based thinking cap, which allows the user to attach various gizmos to the brain. For example, a person with impaired hearing can simply plug in a microphone and use it as an ear.

Currently, corneal transplants are available to restore defective eyes, but in the future, perhaps we will only need to insert artificial eyes with zoom capabilities, infrared sensors and night vision. Cyborg technology will further blur the line between man and machine, in times to come. **EFY**